Sangyeon Cho

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Research Interests

My research interests lie in **causal inference** and **uncertainty quantification** for reliable **decision-making**. I am particularly interested in clarifying the assumptions under which causal conclusions are justified, characterizing uncertainty in causal estimates and predictions, and determining when the available evidence supports action, deferral, or further information. Ultimately, I aim to help bridge the gap between theory and real-world applications in domains such as epidemiology, finance, and public policy.

Keywords: Causal Inference, Uncertainty Quantification, Explainability, Decision-Making, Time-Varying Treatments, Conformal Prediction.

Education

Seoul National University, Seoul, Republic of Korea
Doctor of Philosophy in Data Science **Sep 2026–Present**
Advisor: Sanghack Lee
Master of Data Science **Aug 2026**
Advisor: Sanghack Lee
Thesis: *Conformal Prediction for Individual Treatment Effects under Time-Varying Treatment Strategies*
B.A. in Economics **Aug 2024**

Research Experience

Graduate Student Researcher **Sep 2024–Present**
Causality Lab, Seoul National University

- *Causal Inference–Based Policy Optimization for Determining Unsecured Personal Loan Terms* (AFINIT) 2026–Present

  Project Team Lead

- *AI Platform for Predicting Drug Efficacy and Side Effects from Integrated Medical and Multi-omics Data* (Ministry of Food and Drug Safety) 2026–Present

- *Advancing Credit Decision Making* (AFINIT) 2025–2026

  Project Team Lead: Coordinated research across members while contributing directly to methodology and experiments; the work resulted in the *OPERA-S* manuscript and a Korean patent application and corresponding PCT application (co-inventor).

- *Metabolomic Big Data Analysis* (Ministry of Food and Drug Safety) 2025

  Analyzed Korean cohort multi-omics data through the lenses of causal inference, uncertainty quantification, and explainability, leading to a co-first-authored paper on decomposing conformal uncertainty.

Undergraduate Research Assistant **Jul 2023–Aug 2024**
Causality Lab, Seoul National University

- Developed a research idea titled *Causal Structure and Effect Analysis of South Korea's Low Birth Rate Policy Based on Structural Causal Models*. **Excellence Award, Research Idea Track (2024 CSES Competition)**

Manuscripts

* for equal contribution, † for corresponding authors.

Sangyeon Cho*, Minyoung Cho*, Jungsoo Kim*, Sujeong Oh, and Sanghack Lee†. Decomposing Conformal Uncertainty: Calibration- and Instance-Driven Feature Attribution. In *Advances in Neural Information Processing Systems (NeurIPS 2026)* (to appear).

Sangyeon Cho*, Minyoung Cho*, Jungsoo Kim*, and Sanghack Lee†. A Tale of Two Uncertainties: Global–Local Attribution for Conformal Prediction. *2nd Workshop on Epistemic Intelligence in Machine Learning (EIML), ICML 2026* (non-archival). **Spotlight Talk**

Hyunwoo Kim, Gyeongchan Han, Bogeun Kim, Serjin Kim, **Sangyeon Cho**, Junha Ham, Jaehyeok Shin, and Sanghack Lee†. OPERA-S: Preventing Error Amplification in Ensemble Off-Policy Evaluation. *under review.*

Jaeho Jeong, **Sangyeon Cho**, Ji Won Bae, and Sanghack Lee†. Conformal Meta-learner for Individual Treatment Effects under Unmeasured Confounding. *work in progress.*

Academic Service

Reviewer, Conference on Neural Information Processing Systems (NeurIPS) **2026**

Teaching Experience

Teaching Assistant, Seoul National University

- *Machine Learning and Deep Learning for Data Science II* (Fall 2026)
- *Causal Inference for Data Science* (Spring 2026)
- *Computing Foundations for Data Science* (Spring 2025, Fall 2025; taught two class sessions)

Skills

Languages

- Korean: Native
- English: TOEFL iBT 5.5/6 (110/120 on the previous scale)

Technical Skills

- Programming Languages: Python, C, C++
- Libraries and Frameworks: PyTorch, scikit-learn, NumPy, pandas, Polars

Additional Information

Relevant Coursework
Linear Algebra; Econometrics; Mathematical Foundations of Deep Neural Networks; Computing for Data Science; Machine Learning and Deep Learning for Data Science I–II; Big Data and Knowledge Management System I–II; Data Science Bootcamp, SNU — completed with honors: Linear algebra, vector calculus, probability and statistical inference; Python, data structures, C programming, and computer systems.

Military Service
KATUSA (Korean Augmentation to the United States Army), 2020–2021